Figure 11.2 Inflation can occur repeatedly, sprouting new universes from older ones.

Inflation and the Egg

So why do you see an egg splatter but not unsplatter? Where does the arrow of time that we all experience come from? Here is where this approach has taken us. Through a chance but every so often expectable fluctuation from an unremarkable primordial state with high entropy, a tiny, twenty-pound nugget of space achieved conditions that led to a brief burst of inflationary expansion. The tremendous outward swelling resulted in space's being stretched enormously large and extremely smooth, and, as the burst drew to a close, the inflaton field relinquished its hugely amplified energy by filling space nearly uniformly with matter and radiation. As the inflaton's repulsive gravity diminished, ordinary attractive gravity became dominant. And, as we've seen, attractive gravity exploits tiny inhomogeneities caused by quantum jitters to cause matter to clump, forming galaxies and stars and ultimately leading to the formation of the sun, the earth, the rest of the solar system, and the other features of our observed universe. (As discussed, some 7 billion or so years ATB, repulsive gravity once again became dominant, but this is only relevant on the largest of cosmic scales and has no direct impact on smaller entities like individual galaxies or our solar system, where ordinary attractive gravity still reigns.) The sun's relatively low-entropy energy was used by low-entropy plant and animal life forms on earth to produce yet more low-entropy life forms, slowly raising the total entropy through heat and waste. Ultimately, this chain produced a chicken that produced an egg— and you know the rest of the story: the egg rolled off your kitchen counter and splattered on the floor as part of the universe's relentless drive to higher entropy. It's the low-entropy, highly ordered, uniformly smooth nature of the spatial fabric produced by inflationary stretching that is the analog of having the pages of *War and Peace* all in their proper numerical arrangement; it is this early state of order—the absence of severe bumps or warps or gargantuan black holes—that primed the universe for the subsequent evolution to higher entropy and hence provided the arrow of time we all experience. With our current level of understanding, this is the most complete explanation for time's arrow that has been given.